giving full credence to the concept that the supernovae are associated with stars that have undergone what may be called a final gravitational collapse. If neutron stars can exist for a given range of stellar mass, it is not unreasonable to conclude that black holes will be produced by more massive stars, granting that much of the observational evidence will be indirect. Indeed, there is much such indirect evidence at present.

Finally, several lines of evidence have given additional support to the validity of the general theory of relativity. They include high-precision measurements of spacecraft and planetary orbits in our solar system, and observations of the "lensing" action of some galaxies upon light that reaches us from sources beyond those galaxies. Then, more recently, there is good evidence of the loss of energy of motion of mutually orbiting massive binary stars as a result of the generation of gravitational waves, a major prediction of the theory. Such observations give one courage to believe the untested predictions of the general theory of relativity in the proximity of a black hole and open the path to further imaginative speculation of the type featured here.

Several years ago the Commonwealth Fund decided at the suggestion of its president, Margaret E. Mahoney, to sponsor a Book Program in which working scientists of distinction were invited to write about their work for a literate lay audience. Professor Thorne is such a scientist, and the Book Program is pleased to offer his book as its ninth publication.

The advisory committee for the Commonwealth Fund Book Program, which recommended sponsorship of this book, consisted of the following members: Lewis Thomas, M.D., director; Alexander G. Bearn, M.D., deputy director; Lynn Margulis, Ph.D.; Maclyn McCarty,M.D.; Lady Medawar; Berton Roueché; Frederick Seitz, Ph.D.; and Otto Westphal, M.D. The publisher is represented by Edwin Barber, vice-chairman and editor at W. W. Norton & Company, Inc.

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